

MA1513 Weekly Self Practice Set 3 Answer

1. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 5 & -1 & 1 \\ 2 & 1 & 8 & -2 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 1 & 0 & 3 & -1 & 1 \end{pmatrix}.$$

- (i) Use MATLAB or otherwise to find the reduced row echelon form of $\mathbf{A}$.
- (ii) Use the answer in (i) to find a basis for the row space of $\mathbf{A}$.
- (iii) Find a basis for the column space of $\mathbf{A}$.
- (iv) Find a basis for the nullspace of $\mathbf{A}$.

Answer

$$(i) \text{ rref}(\mathbf{A}) = \begin{pmatrix} 1 & 0 & 3 & -1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

(ii) A basis for the row space of $\mathbf{A}$ is $\{(1, 0, 3, -1, 0), (0, 1, 2, 0, 0), (0, 0, 0, 0, 1)\}$

(iii) A basis for the column space of $\mathbf{A}$ is

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

(iv) To find the nullspace of $\mathbf{A}$, we solve the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$:

Let the 5 variables of the system be x_1, x_2, x_3, x_4, x_5 . Then using the rref of $\mathbf{A}$ in (i), we have:

$$x_3 = s, x_4 = t, x_5 = 0, x_2 = -2x_3 = -2s, x_1 = -3x_3 + x_4 = -3s + t.$$

This gives the general solution:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -3s + t \\ -2s \\ s \\ t \\ 0 \end{pmatrix} = s \begin{pmatrix} -3 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

and hence a basis for the nullspace is given by

$$\left\{ \begin{pmatrix} -3 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

2. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be a subset of $\mathbb{R}^6$ where

$$\mathbf{v}_1 = (1, 2, -2, 1, 1, 0), \mathbf{v}_2 = (-2, -4, 4, -2, -2, 0), \mathbf{v}_3 = (2, 1, 1, 2, 0, 1),$$

$$\mathbf{v}_4 = (1, -1, 3, 1, -1, 1).$$

Let $V = \text{span}(S)$, a subspace of $\mathbb{R}^6$.

- (i) Is S a linearly independent set? Why?
- (ii) Find a basis for V that is made up of (some of) the vectors in S . (Refer to section 2.5.)
- (iii) What is the dimension of V ?

Answer

- (i) We use the 4 vectors (column form) to form a 6×4 matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 2 & 1 \\ 2 & -4 & 1 & -1 \\ -2 & 4 & 1 & 3 \\ 1 & -2 & 2 & 1 \\ 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

and reduce it (using MATLAB) to rref:

$$\begin{pmatrix} 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since not all columns in the rref are pivot columns, S is linearly dependent.

- (ii) To find a basis for V from the set S , we take the first and third columns of $\mathbf{A}$ that correspond to the pivot columns of the rref. This gives $\{(1, 2, -2, 1, 1, 0), (2, 1, 1, 2, 0, 1)\}$ as a basis for V .
- (iii) From (ii), the dimension of V is 2.

3. (Use MATLAB with format rat option to find the answers of this question.)

(i) Let $\mathbf{A} = \begin{pmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \\ 40 & 41 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 20 \\ 40 \\ 30 \\ 10 \end{pmatrix}$. Find $\mathbf{A}^T \mathbf{A}$ and $\mathbf{A}^T \mathbf{b}$.

(ii) Use (i) to find the least squares solution to the system $\mathbf{A}\mathbf{x} = \mathbf{b}$.

(iii) Use (ii) to find the projection of the vector $(20, 40, 30, 10)$ onto the subspace given by $\text{span}\{(10, 20, 30, 40), (11, 21, 31, 41)\}$. (See section 2.7)

(Give your answers for (ii) and (iii) in fractions)

Answer

(i) $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 3000 & 3100 \\ 3100 & 3204 \end{pmatrix}$ and $\mathbf{A}^T \mathbf{b} = \begin{pmatrix} 2300 \\ 2400 \end{pmatrix}$

(ii) The least squares solution of this system is given by

$$\mathbf{v} = \begin{pmatrix} 3000 & 3100 \\ 3100 & 3204 \end{pmatrix}^{-1} \begin{pmatrix} 2300 \\ 2400 \end{pmatrix} = \begin{pmatrix} -\frac{177}{5} \\ 35 \end{pmatrix}$$

(iii) The projection is given by

$$\mathbf{p} = \mathbf{A}\mathbf{v} = \begin{pmatrix} 31 \\ 27 \\ 23 \\ 19 \end{pmatrix}$$